DIPLOMA PROGRAMME PRACTICE · HL · SET A

Biology / Paper 2

Question booklet · 80 marks · 150 minutes

Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams. Answer all of Section A and two of three Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY

QUESTION 1 [12]

Heat exposure and a heritable shell trait

Researchers reared snails from several families under common conditions. Within each colour morph, siblings were randomly allocated to cool and hot chambers. Each treatment used five independently regulated chambers. Counts are pooled across chambers. All snails were juveniles at the start.

Morph	Initial per temperature	Survived cool	Survived hot
Light	100	85	72
Dark	100	88	48

(a) Calculate the survival reduction caused by heat for each morph in percentage points and compare the reductions. **[3]**

(b) Assuming equal starting numbers, determine the light-morph frequency among hot-chamber survivors and explain why it is not an allele frequency. **[3]**

(c) Explain why dividing siblings between treatments strengthens a temperature comparison and why individual snails do not all represent independent temperature replicates. **[3]**

(d) Evaluate the claim that the next generation in the wild must have 60% light individuals. **[3]**

SECTION A · COMPULSORY
QUESTION 2 [12]

Testing a mitochondrial inhibitor

Isolated mitochondria are supplied with respiratory substrate, ADP, phosphate and oxygen. At the indicated time a compound is added. Its proposed action is to make the inner membrane permeable to protons. Measurements are steady rates or relative values under each condition.

Condition	O_2 uptake / nmol min^{-1}	ATP synthesis / nmol min^{-1}	Relative proton gradient
Control	24	96	1.00
Compound	30	30	0.35

(a) Calculate ATP synthesized per molecule of O_2 consumed under each condition and the percentage decrease in this ratio. **[3]**

(b) Explain the relationship between electron transport and the proton gradient under control conditions. **[3]**

(c) Explain how a proton leak could produce all three observed changes. **[3]**

(d) A second compound stops all electron transfer to oxygen. Predict two outcomes and identify one useful control when testing these compounds. **[3]**

SECTION A · COMPULSORY
QUESTION 3 [12]

Tracking parental DNA

Bacteria grown for many generations with heavy nitrogen are moved to a light nitrogen supply. DNA bands are separated by density. There is one chromosome per starting cell and synchronous replication. The expected bands below refer to intact double-stranded molecules.

Time	Heavy band	Intermediate band	Light band
Before transfer	100%	0%	0%
After one round	0%	100%	0%
After two rounds	0%	50%	50%

(a) Use strand inheritance to explain the intermediate band after one round. **[3]**

(b) Predict the percentages following a third round and justify your prediction. **[3]**

(c) Explain why observations from both rounds distinguish semiconservative replication from a simple dispersive model that mixes old and new segments in every daughter strand. **[3]**

(d) Explain why the two new strands at one replication fork are synthesized differently despite complementary base pairing on both templates. **[3]**

SECTION A · COMPULSORY
QUESTION 4 [12]

Water conservation after exercise

Volunteers complete a controlled exercise session. Over a fixed subsequent period group A drinks water and group B receives a smaller equal-temperature drink. The data are group means. No solute is added to either drink.

Measure	Group A	Group B
Plasma osmolarity / mOsm L^{-1}	287	299
Relative ADH concentration	1.0	2.4
Urine volume per hour / cm^3	180	75
Urine osmolarity / mOsm L^{-1}	310	640

(a) Calculate approximate urinary osmole excretion in mOsm h^{-1} in each group and compare it with the difference in urine volume. **[3]**

(b) Explain the chain from increased plasma osmolarity to increased collecting-duct water permeability. **[3]**

(c) Explain how that permeability change leads to group B's urine pattern. **[3]**

(d) Evaluate the claim that the drink volume alone caused the entire difference in ADH. **[3]**

SECTION B · EXTENDED RESPONSE
QUESTION 5 [16]

From motor neuron to shortening muscle

Consider transmission at a neuromuscular junction and contraction of a skeletal muscle fibre.

(a) Outline how the electrical signal initiates chemical transmission at the motor neuron terminal. **[3]**

(b) Explain the sequence from transmitter binding to exposure of myosin-binding sites on actin. **[5]**

(c) Explain cross-bridge cycling and how its microscopic effects produce shortening and relaxation. **[7]**

[1] Overall quality of the extended response.

SECTION B · EXTENDED RESPONSE
QUESTION 6 [16]

The same genome, different cells

Two tissues from one individual synthesize different amounts and variants of a membrane receptor. Their receptor-gene coding DNA is identical.

(a) Distinguish genome, transcriptome and proteome in these tissues. **[3]**

(b) Explain how interactions at DNA can produce different transcription rates in the two tissues. **[5]**

(c) Explain how RNA processing and subsequent events can affect the form and abundance of functional receptor present at the cell surface. **[7]**

[1] Overall quality of the extended response.

SECTION B · EXTENDED RESPONSE
QUESTION 7 [16]

Evolution in a changing population

A mainland insect population is large. A few insects establish a population on an island with a new predator and reduced immigration.

(a) Outline the assumptions directly needed for random mating to produce Hardy–Weinberg genotype proportions at a two-allele autosomal locus. **[3]**

(b) Explain how founding and genetic drift can change allele frequencies without a fitness difference. **[5]**

(c) Explain how selection and reduced gene flow could contribute to speciation, and distinguish evidence of divergence from evidence of reproductive isolation. **[7]**

[1] Overall quality of the extended response.